

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: Second Year B. Tech (Sem-IV) Branch : Civil Engineering Semester :IV

Subject Code & Name: BTCVC404/BTCIC404/BTCEC404 Water Resources Engineering

Max Marks: 60 Date: 27/12/2024 Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	What is the term used to describe the relationship between the area irrigated and the volume of water used?	BL2/CO 1	1
	a) Base period b) Delta c) Duty d) Consumptive use		
2	Which type of irrigation system involves delivering water using gravitational flow?	BL2/CO 1	1
	a) Drip irrigation b) Sprinkler irrigation c) Canal irrigation d) Lift irrigation		
3	What does a mass curve represent in reservoir planning?	BL2/CO 1	1
	a) Rainfall data b) Cumulative inflow and outflow c) Flood frequency d) Water losses		
4	The useful life of a reservoir is primarily reduced by:	BL2/CO 1	1
	a) Silting b) Rainfall c) Evaporation losses d) Demand curve		
5	Which of the following is a primary function of a gallery in a gravity dam?	BL2/CO 2	1
	a) Enhancing stability b) Preventing seepage c) Drainage and inspection d) Increasing water pressure		
6	Which failure mode is specific to earth dams?	BL2/CO 1	1
	a) Sliding b) Piping c) Overturning d) Arching		
7	The Bligh's creep theory is used to analyze:	BL2/CO 2	1
	a) Silt deposition b) Seepage in weirs c) Stability of dams d) Alignment of canals		
8	What is the primary function of inverted filters in earth dams?	BL2/CO 1	1
	a) To increase drainage efficiency b) To reduce uplift pressure c) To minimize seepage d) To support structural stability		
9	The unit hydrograph is used to determine:	BL2/CO 2	1
	a) Rainfall intensity b) Catchment area size c) Runoff response from rainfall d) Base flow discharge		
10	Which formula is typically used to estimate runoff in a catchment?	BL2/CO 2	1

	a) Darcy's equation	b) Manning's formula	c) Rational method	d) Bernoulli's equation		
11	Flood hydrographs are primarily used to study:				BL2/CO 3	1
	a) Evaporation losses	b) Rainfall distribution	c) Flow variations during a flood	d) Groundwater recharge		
12	Which type of reservoir classification is based on its purpose?				BL2/CO3	1
	a) Single-purpose	b) Zoned reservoir reservoir	c) Overflow reservoir	d) Silting reservoir		
Q. 2 Solve the following.						12
A)	Derive relationship between duty and delta.				BL2/CO 1	6
B)	Differentiate between the lift irrigation and flow irrigation				BL2/CO1	6
Q.3 Solve the following.						12
A)	Explain the process of selecting a site for a reservoir. What are the key factors that influence the site selection for constructing a reservoir?				BL3/CO2	6
B)	Explain elementary profile of gravity dam.				BL3/CO1	6
Q. 4 Solve Any Two of the following.						12
A)	Discuss the components of an earthen dam, their functions, and the design criteria to ensure stability and prevent failure.				BL3/CO2	6
B)	Explain Bligh's creep theory and state its limitations.				BL2/CO2	6
C)	Determine the section of Lacey's regime channel to carry 15 m ³ /s of water in an alluvium with a sediment size of 0.75 mm.				BL4/CO3	6
Q.5 Solve Any Two of the following.						12
A)	Explain the hydrologic cycle and describe the major processes involved in it.				BL2/CO2	6
B)	What is a flood hydrograph, and how is it used to estimate flood flow in a river or stream?				BL2/CO2	6
C)	In a catchment area, there are five existing rain gauge stations. The annual rainfall at these stations is recorded as follows: 800 mm, 720 mm, 600 mm, 850 mm, and 780 mm. The allowable percentage error in the mean rainfall value is 12%. Calculate the optimum number of rain gauges required in the catchment to achieve the desired accuracy.				BL2/CO1	6
Q. 6 Solve Any Two of the following.						12
A)	Describe the causes of water logging in irrigated lands. Discuss the preventive and curative measures for it.				BL2/CO3	6
B)	Explain with neat sketch rain water harvesting.				BL3/CO3	6
C)	A tube well is constructed in a basin with a specific yield of 0.25. If the area of influence of the well is 1000 m ² and the drawdown during pumping is 10 m, calculate the discharge of the tube well. Use Darcy's Law for the calculation.				BL4/CO3	6